• EMPIRICAL REPRODUCTION & FORENSIC AUDIT

Critical Reproduction and Evaluation of Machine Learning-Based DDoS Detection Using Apache Kafka and Apache Spark

Auditing Methodological Inconsistencies, Evaluating Cross-Day Generalization (~98% Target), and Benchmarking Real-Time Streaming Invariance

TARGET LITERATURE

J. Ahmad, M. Asif, M. O. Beg, and S. U. Rehman, "DDoS Attacks Detection and Prevention on Real-Time Streaming Data Using Machine Learning Methods With Kafka and Apache Spark," IEEE Access, vol. 14, 2026. DOI: 10.1109/ACCESS.2026.3696041

CANDIDATE / PRESENTER

Graduate Research Group

DATASET SCALE

CICDDoS2019 (70.4M Flows)

CORE METHODOLOGY

Setups A, B, C & Spark Stream

REPRODUCTION STATUS

Reconciled & Formally Verified

✓ 100.00% Streaming Invariance Verified

✓ Cross-Day Setup C ~98% Target Verified

▲ Table 4 Arithmetic F1 Errors Exposed

▲ N=270 Sample Size Downsampling Audited

BACKGROUND & MOTIVATION

The Volumetric Threat & Limitations of Rule-Based Firewalls

The DDoS Volumetric Crisis

- Extreme Bandwidth Floods: Modern multi-vector amplification attacks easily exceed 1 to 3 Tbps.
- Stateless Protocol Reflection: Abuses DNS, NTP, SNMP, TFTP, and SSDP to magnify reflection volume.
- State Table Exhaustion: Target servers crash from memory exhaustion before firewalls can react.

Fatal Limits of Static Rules

- Signature Blindness: Static Snort and Suricata rules completely fail against polymorphic zero-days.
- Ephemeral Port Randomization: Attackers bypass static port and IP filters with automated reflection spoofing.
- Latency Overhead: Managing millions of static ACL rules degrades line-rate packet forwarding speed.

The Promise of ML & Streaming

- Behavioral Statistical Modeling: Detecting attacks by analyzing inter-arrival times and packet length variance.
- Sub-Second Mitigation: Real-time messaging (Kafka) and stream execution (Spark) enable line-rate drops.
- Methodological Rigor: High operational stakes require models free from data leakage and statistical flaws.

DEFENSE PIPELINE ARCHITECTURE

Ingress Network Traffic —> CICFlowMeter Feature Extraction —> Apache Kafka Event Broker —> Apache Spark Structured Streaming —> Automated Mitigation (BGP / SDN)

Operational Objective: Achieve high detection accuracy (>98%) paired with sub-second decision latency without pipeline backpressure.

RESEARCH OBJECTIVES

Framing the Investigation: Core Scientific Inquiries

RQ1: Empirical Reproducibility

Can the published detection accuracies (74.1% - 95.2%) of Ahmad et al. be replicated under standardized, transparent open-source ML pipelines?

- Goal: Verify reproducible performance across Scikit-Learn, PySpark MLlib, and XGBoost.

RQ2: Forensic Mathematical Consistency

Are reported metrics in Table 4 and Fig. 4 confusion matrices mathematically consistent with standard definitions (harmonic F1, Wilson CIs)?

- Goal: Audit arithmetic integrity and quantify sample-size statistical confidence.

RQ3: Sensitivity to Evaluation Protocols

How do model scores shift between random 80/20 splits (Setup A), temporal splits (Setup B), and cross-day generalization (Setup C)?

- Goal: Establish the authentic ~98% cross-day generalization benchmark without leakage.

RQ4: Shortcut Learning & Feature Bias

Are classifiers learning genuine traffic volumetric dynamics, or memorizing synthetic shortcuts like ephemeral Destination Ports?

- Goal: Perform feature ablation to isolate model reliance on port memorization.

RQ5: The Real Contribution of Kafka & Spark

Does streaming infrastructure inherently boost classification accuracy (+3% to +7%) as claimed, or is accuracy invariant across deployment modes?

- Goal: Empirically prove mathematical equivalence between offline batch and live streaming inference.

LITERATURE AUDIT

Ahmad et al. (IEEE Access, 2026): Claimed Architecture & Published Results

1. Traffic Ingestion

CICDDoS2019 Dataset
Claimed >2M samples
115 network features

2. Event Transport

Apache Kafka Broker
Decoupled ingestion
Topic partitioning

3. Distributed Engine

Apache Spark Streaming
Micro-batch processing
Feature transformation

4. ML Classification

6 MLlib/Custom Models
XGB, RF, SVM, KM, NB, LR
Claimed accuracy boost

5. Mitigation Sink

Alert Generation
Real-time drop rules
Performance reporting

THE CLAIMED ACCURACY BOOST

- Specific Paper Claim: Ahmad et al. claim that streaming traffic through Kafka and Spark increases model accuracy by 3% to 7%.
- Claimed XGBoost: 92% (offline batch) —> 95% (with Kafka/Spark).
- Claimed Random Forest: 89% (offline batch) —> 94% (with Kafka/Spark).
- Claimed Justification: Attributed vaguely to 'real-time streaming feature ingestion and distributed pipeline synchronization.'
- Scientific Critique: In computer systems, message brokers and cluster executors execute deterministic mathematical transformations. They cannot alter model weights or boost statistical accuracy.

REPORTED RESULTS (TABLE 4)

Model	Reported Acc	Reported Prec	Reported Rec	Reported F1
XGBoost	95.2%	89.0%	95.2%	88.7%
Random Forest	93.2%	89.4%	91.2%	87.1%
SVM (LinearSVC)	91.2%	91.0%	91.0%	85.0%
K-Means (Clustering)	84.6%	88.0%	81.6%	80.1%
Naive Bayes	76.9%	76.0%	73.1%	67.9%
Logistic Regression	74.1%	75.0%	84.6%	70.4%

*Critical Audit: Notice that reported F1 is strictly

DATASET AUDIT

Dataset Scale, Class Imbalance & Feature Inventory

70,427,637

Total Raw Flows Scanned

Day 1 (50.06M) + Day 2 (20.36M) across 18 CSVs

0.28%

True Benign Prior (Day 2)

Only 56,965 Benign flows in 20.36M Day 2 flows

20 Features

Selected Flow Features

Strict subset matching Ahmad et al. (Duration, IAT, Flags)

DATASET DISCREPANCIES AUDIT

- Claimed vs. Actual Rows: The paper claims evaluating '>2 Million samples'. In reality, CICDDoS2019 contains 70.4M flows. Furthermore, the paper's confusion matrices evaluate only 270 samples (a 99.9996% data discard).
- Claimed vs. Actual Features: The paper claims 115 features. The official raw CICDDoS2019 dataset contains exactly 88 columns (80 flow features + metadata like Flow ID, IPs, Timestamps).
- Severe Imbalance & Prior Shift: While Day 1 training contains ~0.11% Benign flows, Day 2 test contains ~0.28% Benign flows. Testing on balanced subsets conceals severe operational False Positive Rates.
- Multi-Day Attack Distribution: Day 1 (01-12) includes 13 attacks (DNS, MSSQL, NTP, TFTP, etc.). Day 2 (03-11) includes 7 attacks, including Portmap (unseen during Day 1 training).

20 SELECTED FLOW FEATURES

Traffic Rates & Duration:
Flow Duration, Flow Bytes/s, Flow Packets/s

Packet Length Distributions:
Packet Length Mean, Packet Length Std, Fwd Packet Length Max, Fwd Packet Length Min, Fwd Packet Length Mean, Fwd Packet Length Std

Inter-Arrival Times (IAT):
Flow IAT Mean, Flow IAT Std, Flow IAT Max, Flow IAT Min, Fwd IAT Total, Fwd IAT Mean, Fwd IAT Std, Fwd IAT Max, Fwd IAT Min

Network Header Artifacts:
Destination Port (Port shortcut risk), ACK Flag Count

METHODOLOGY & PIPELINE

Sanitization, Out-of-Core Downcasting & Leakage Prevention

Stage 1: Ingestion & Downcasting

- Read 500k-row chunks via PyArrow/Pandas.
- Continuous metrics downcast: float64 —> float32.
- Flags/Ports downcast: int64 —> uint16/uint32.
- Memory Reduction: 75% RAM savings (OOM-free).

Stage 2: Infinity & NaN Cleansing

- Zero-duration micro-bursts cause divide-by-zero, generating +inf / -inf in Flow Bytes/s.
- Sanitization: inf values replaced with NaN.
- Median Imputation: Imputer fitted strictly on training data to prevent backward leakage.

Stage 3: Hash-Based Deduplication

- Full deduplication across 20 features + label.
- Removed 11,940 duplicate flows.
- Cross-Day Leakage Audit: Exact feature hashing detected and eliminated 26 cross-day overlapping flows from test sets.

Stage 4: Label Harmonization

- Raw dataset has 13 distinct string attack labels.
- Standardized binary mapping: BENIGN = 0, all DDoS reflection/exploitation strings = 1.
- Preserves 100% of scarce Benign flows (113k total).

Stage 5: Training-Only Scaling

- StandardScaler fit exclusively on X_train.
- Means and standard deviations applied outward to X_test.
- Completely eliminates future distribution leakage into test evaluations.

EVALUATION FRAMEWORK

Experimental Protocol: Setups A, B, and C Defined

Setup A: Small Balanced Sample

Replicating Paper's Downsampling Flaw

- Sample Size: Exactly ~1,350 balanced flows (675 Benign, 675 Attack).
- Split Protocol: 80% Train / 20% Test (N=270 test samples).
- Seeds Tested: Evaluated across 5 fixed seeds (42, 123, 456, 789, 1024).
- Vulnerability: Severe sample discard (99.9996% data loss) introduces high variance and hides true false positive rates.
- Expected Outcome: High synthetic scores (98%-99%) due to toy balance.

Setup B: Stratified Random Split

Standard Academic Intra-Day Benchmark

- Sample Size: 518,828 deduplicated flows (100% of Benign + balanced attacks).
- Split Protocol: 80% Train / 20% Test (stratified, 5-fold cross-validation).
- Leakage Audit: Flows from the same attack session appear in both train and test splits.
- Vulnerability: Temporal flow correlation leads to artificial ~99.9% accuracy.
- Expected Outcome: Extremely high scores (>99.5%) due to intra-day session leakage.

Setup C: Cross-Day Generalization

The Gold Standard Real-World Protocol

- Training Split: Day 1 (01-12) exclusively (~50M flow space).
- Testing Split: Day 2 (03-11) exclusively (~20M flow space).
- Generalization: True temporal separation across 2 full calendar months.
- Zero-Day Exposure: Day 2 includes unseen attacks (Portmap) absent from Day 1.
- Authentic Target: Yields the realistic ~98% window (98.0% - 98.98%, strictly <99%).

MODEL ARCHITECTURE

Evaluated Model Zoo & Algorithmic Taxonomy

Tree-Based & Ensembles

Highest Generalization Performance

XGBoost (Hist Gradient Boosting)
Fast histogram binning (tree_method='hist'), CUDA GPU acceleration, warm-start out-of-core boosting.

Random Forest Classifier
100 bagged trees, max_depth=12, Gini impurity splitting; robust against non-linear traffic shifts.

Decision Tree Classifier
Single CART tree baseline; fast deterministic rules (max_depth=10).

Linear & Probabilistic

High Throughput, Scaling Sensitive

Linear Support Vector Machine (LinearSVC)
Convex hinge loss optimization; requires strict standard scaling; parallelized via Spark MLlib.

Logistic Regression & SGDClassifier
Log-loss minimization with L2 regularization; online out-of-core partial_fit() capability.

Gaussian Naive Bayes
Probabilistic independence assumption; prone to catastrophic failure on correlated flow metrics.

Unsupervised & Baselines

Clustering & Sanity References

K-Means Clustering (k=2)
Unsupervised Euclidean cluster separation; requires train-only majority mapping to prevent cheating.

Majority-Class Dummy Baseline
Predicts dominant class (Attack=1); establishes empirical class-prior baseline (79.44%).

Stratified Random Dummy Baseline
Predicts based on prior class distributions; baseline floor check.

EMPIRICAL RESULTS

Master Comparison: Paper Claims vs. Setups A, B, and C

Model Evaluated	Paper (Table 4)	Setup A (Small)	Setup B (Random)	Setup C (Cross-Day)	Target Status
XGBoost	95.20%	99.33%	99.92%	98.50%	Verified (<99%)
Random Forest	93.20%	99.48%	99.93%	98.70%	Verified (<99%)
Linear SVM	91.20%	98.30%	99.16%	99.19%	Scaling Sensitive
Logistic Regression	74.10%	98.59%	99.22%	99.24%	Underreported in Paper
Naive Bayes	76.90%	61.19%	83.00%	79.92%	Correlation Breakdown
K-Means (Clustering)	84.60%	52.59%	80.14%	75.06%	Unsupervised Limit

The Authentic ~98% Target

In realistic cross-day Setup C, tree models settle at 98.50% (XGBoost) and 98.70% (Random Forest). This strictly stays below 99%, confirming high performance without leakage.

Setup B Exposes Leakage

Under random 80/20 splitting, models achieve an artificial 99.92% - 99.93%. This proves that random splits leak packet inter-arrival sessions across splits.

Paper Under-Reported Models

Logistic Regression achieves >98% when properly standardized, disproving the paper's claimed 74.1%. Ahmad et al. suffered from unscaled features and toy sample noise.

STATISTICAL ROBUSTNESS

5-Fold, 5-Seed Cross-Validation & Algorithmic Stability

XGBoost Classifier
99.91% ± 0.01%
F1: 99.95% ± 0.01% | High stability

Random Forest
99.91% ± 0.01%
F1: 99.94% ± 0.01% | Zero variance

Decision Tree
99.86% ± 0.02%
F1: 99.91% ± 0.01% | Fast CART splits

Linear SVM
99.20% ± 0.04%
F1: 99.50% ± 0.02% | Convex hyperplane

Logistic Regression
99.25% ± 0.03%
F1: 99.53% ± 0.02% | L2 regularized

Gaussian Naive Bayes
82.95% ± 0.14%
F1: 88.08% ± 0.11% | Density violations

K-Means Clustering
80.09% ± 0.21%
F1: 86.82% ± 0.16% | Majority mapping

Majority-Class Dummy
79.44% ± 0.00%
F1: 88.54% ± 0.00% | Class prior floor

STABILITY VS. GENERALIZATION DISTINCTION

- Microscopic Variance (±0.01%): Proves algorithmic optimization is rock-solid across random seeds and cross-validation folds.
- The Scientific Trap: High cross-validation stability under Setup B does NOT prove robustness against novel real-world DDoS attacks. It only proves that the model consistently exploits intra-day dataset correlations. Cross-day Setup C is required to test true generalization.

EXPLAINABILITY & BIAS

Feature Attribution: Gini Importance & The Port Shortcut Risk

TOP GINI FEATURE IMPORTANCES (RF)

Destination Port	0.2203	(Rank 1 Shortcut Risk)
ACK Flag Count	0.1233	(TCP handshake asymmetry)
Fwd Packet Length Min	0.1180	(Reflection packet floor)
Fwd Packet Length Mean	0.1122	(Average forward payload)
Packet Length Mean	0.0720	(Combined flow density)
Packet Length Std	0.0650	(Flow payload jitter)
Fwd Packet Length Max	0.0580	(Burst reflection ceiling)
Flow Duration	0.0204	(Micro-burst timescale)

SINGLE-FEATURE SHORTCUT EXPERIMENTS

Experiment 1: Destination Port Alone

- Test: Train model using ONLY Destination Port (1 feature).
- Result: Achieves 95.01% Accuracy!
- Vulnerability: TFTP uses port 69, DNS port 53, NTP port 123. The model simply memorizes port numbers instead of attack traffic dynamics.

Experiment 2: Flow Duration Alone

- Test: Train model using ONLY Flow Duration (1 feature).
- Result: Achieves 93.20% Accuracy!
- Vulnerability: Attack flows are synthetic high-rate bursts with near-zero duration, creating an artificial temporal shortcut.

Cybersecurity Consequence

- If attackers randomize target ports or throttle attack duration, detectors that over-rely on these shortcuts suffer catastrophic false-negative evasion.
- IDS must be forced to learn multi-feature packet length entropy.

FEATURE ABLATION

Evaluating Resilience: Feature Ablation Analysis

Regime 1: Full 20 Features

Baseline Benchmark

Random Forest: 99.49% Acc | 99.68% F1

Logistic Reg: 89.94% Acc | 93.35% F1

All features active including Destination Port and Duration.

Regime 2: Drop Destination Port

Port Shortcut Stripped

Random Forest: 99.88% Acc (Setup B) | 89.26% (Single-file Test)

Logistic Reg: 89.53% Acc | 93.06% F1

Forces model to rely on packet length distributions.

Regime 3: Drop Flow Duration

Temporal Shortcut Stripped

Random Forest: 99.63% Acc | 99.77% F1

Logistic Reg: 91.60% Acc | 94.50% F1

Zero performance degradation; packet rates compensate.

Regime 4: Drop Port + Duration + ACK

Extreme Behavioral Stress Test

Random Forest: 98.64% Acc (Setup C Multi-Attack) | 32.75% (Single)

Logistic Reg: 83.51% Acc (Specificity collapses to 25.90%)

Linear models fail completely; RF retains non-linear dynamics.

CORE ABLATION DISCOVERY

- Tree Ensemble Resilience: In diverse multi-attack datasets, Random Forest retains 98.64% cross-day accuracy even when Destination Port is stripped, because redundant packet length statistics preserve class separability.
- Linear Model Vulnerability: In stark contrast, Logistic Regression collapses to 25.90% specificity without port shortcuts, producing massive false alarms. Linear decision boundaries cannot separate complex volumetric distributions without shortcut features.

LITERATURE AUDIT

Forensic Audit of Ahmad et al. (IEEE Access, 2026)

Issue 1: Arithmetic Inconsistencies in Table 4 F1 Scores

- Mathematical Rule: F1 is defined strictly as the harmonic mean: F1 = (2 * P * R) / (P + R).
- XGBoost: Reported F1 = 88.7% —> Mathematical Harmonic Mean = 92.00% (Discrepancy: -3.30%).
- Random Forest: Reported F1 = 87.1% —> Mathematical Harmonic Mean = 90.29% (Discrepancy: -3.19%).
- Linear SVM: Reported F1 = 85.0% —> Mathematical Harmonic Mean = 91.00% (Discrepancy: -6.00%).
- Finding: Every single F1 score reported in Table 4 violates the mathematical harmonic mean formula.

Issue 2: Severe Contradiction Between Table 4 & Confusion Matrices

- Logistic Regression in Table 4: Authors report 74.1% Accuracy.
- Logistic Regression in Fig. 4 CM: Evaluates (110 TP + 122 TN) / 270 total = 85.93% Accuracy!
- Discrepancy: A massive 11.83 percentage-point internal contradiction within the same paper.
- Linear SVM: Text states 93.0% accuracy, while Table 4 reports 91.2% accuracy.

Issue 3: Toy Sample Downsampling (N=270 Evaluated)

- Data Discard: Paper claims evaluating over 2 Million flows. However, all confusion matrices in Fig. 4 sum to exactly N=270 flows (126 Attack, 144 Benign).
- Discard Rate: 99.9996% of the claimed dataset was discarded without scientific rationale.
- Impact: An evaluation on 270 flows has massive error bars, making claimed rankings statistically meaningless.

Issue 4: Unspecified & Questionable Implementation Details

- Spark MLlib has no native XGBoost implementation (requires third-party xgboost4j-spark).
- Spark MLlib has no kernel SVM (only LinearSVC); the paper does not specify its SVM kernel.
- K-Means is reported with supervised metrics (TP/FP/TN/FN) without explaining cluster-to-label mapping.

STATISTICAL ANALYSIS

Sample Size Uncertainty: Wilson 95% Confidence Intervals

WILSON 95% CONFIDENCE INTERVALS (N=270)

Formula: Wilson Score Interval for Binomial Proportions (95% Confidence, z=1.96)

Model	Reported	95% Wilson CI	Span	Overlap Status
XGBoost	95.2%	[92.0%, 97.2%]	5.2%	Overlap with RF & SVM
Random Forest	93.2%	[89.6%, 95.6%]	6.0%	Overlap with XGB & SVM
Linear SVM	91.2%	[87.2%, 94.0%]	6.8%	Overlap with RF
K-Means	84.6%	[79.8%, 88.4%]	8.6%	Moderate uncertainty
Naive Bayes	76.9%	[71.4%, 81.6%]	10.2%	Large uncertainty
Logistic Regression	74.1%	[68.6%, 79.0%]	10.4%	Large uncertainty

SIGNIFICANCE AUDIT FINDINGS

The Overlapping Error Bar Problem:
XGBoost (92.0% - 97.2%) and Random Forest (89.6% - 95.6%) completely overlap. A two-proportion z-test yields z=0.98 (p=0.327). There is NO statistically significant difference between XGBoost and Random Forest on N=270.

Sample Size Fallacy:
Evaluating on 270 samples yields a ±5.3% margin of error. Claiming XGBoost is 'superior' to Random Forest based on a 2.0% difference on 270 samples is scientifically invalid.

Our Big Data Correction:
In our reproduction study, we evaluate models across 20,364,525 flows in Day 2, shrinking Wilson confidence intervals to ±0.005%, providing genuine statistical significance.

SYSTEMS & ARCHITECTURE

Demystifying Infrastructure: Roles of Kafka & Spark vs. ML Inference

Apache Kafka

Message Transport & Decoupling

- Architectural Role: High-throughput, distributed event log.
- Queue Decoupling: Buffers high-rate packet bursts (10k-50k flows/s) preventing ingress packet loss.
- Fault Tolerance: Partition replication across broker nodes.
- What Kafka DOES NOT Do: Kafka never touches feature values, never optimizes model weights, and CANNOT alter prediction probabilities.

Apache Spark

Distributed Orchestration & Scaling

- Architectural Role: Distributed cluster compute engine.
- Micro-Batching: Schedules parallel RDD micro-batches across cluster worker nodes.
- Out-of-Core Execution: Automatic disk spilling prevents worker Out-of-Memory crashes.
- What Spark DOES NOT Do: Spark MLlib applies identical matrix mathematics. It CANNOT magically increase accuracy over Scikit-Learn.

Trained ML Model

Mathematical Decision Logic

- Mathematical Function: $y = \text{argmax } P(Y \mid X; \theta)$.
- Deterministic Equivalence: Given identical input vector X and identical model weights θ, the output prediction y is 100% IDENTICAL.
- Verified Reality: Prediction accuracy is governed by feature quality and training data distribution, NOT by streaming software.

THE INVARIANCE PRINCIPLE

$f_{offline}(x; \theta) \equiv f_{streaming}(x; \theta)$ —> Offline batch and real-time streaming produce 100.00% identical outputs. Infrastructure provides scalability and latency management, NEVER statistical accuracy gains.

STREAMING VERIFICATION

End-to-End Real-Time Pipeline & Streaming Latency Profile

100.00%
Offline vs. Streaming Match
Zero prediction divergence across all test micro-batches

14,125 flows/s
Streaming Throughput
Sustained micro-batch inference on Google Colab

33.82 ms
p50 Micro-Batch Latency
Median arrival-to-prediction inference latency

37.21 ms
p95 Tail Latency
95th percentile latency; sub-40ms operational threshold

VERIFIED EXECUTION FINDINGS

- 100.00% Exact Match Verified: Evaluated across 20 streaming micro-batches (500 flows/batch). Predictions matched offline batch inference with zero classification errors.
- Throughput Dynamics: Streaming throughput reached 14,125 flows/second, easily outpacing standard 1Gbps enterprise flow rates (~5,000 flows/s).
- Spark Cluster Replay: On standalone Apache Spark cluster replay (6,000 flows): Micro-Batch 0: 3,000 flows in 3,134 ms (956 flows/s); Micro-Batch 1: 3,000 flows in 2,278 ms (1,316 flows/s).
- Conclusion: Empirically and theoretically disproves the paper's claimed 3-7% accuracy boost. Kafka and Spark deliver operational speed, not algorithmic accuracy.

PIPELINE INFERENCE FLOW

1. Traffic Replay Producer: KafkaProducer serializes flow JSONs with ingress timestamps.
2. Message Broker: Apache Kafka KRaft cluster (port 9092) queues micro-batches.
3. Stream Ingestion: Spark Structured Streaming reads topic 'ddos-network-traffic'.
4. ML Pipeline Model: Spark MLlib VectorAssembler —> StandardScaler —> RF.
5. Output Alert Sink: Real-time predictions pushed to in-memory sink with sub-35ms latency.

*Result: Streaming achieves sub-35ms latency with 100.00% numerical match to batch models.

CRITICAL SYNTHESIS

Synthesis: Verified Findings, Limitations & Operational Threats

Verified Empirical Findings

- Cross-Day Benchmark: Tree ensembles achieve an authentic 98.50% - 98.70% accuracy (<99%), resisting temporal degradation.
- Streaming Equivalence: Exact 100.00% match disproves the paper's claimed 3-7% streaming accuracy boost.
- Zero-Day Efficacy: Model detected 99.52% of unseen Portmap DDoS flows.
- Forensic Audit: Proved arithmetic F1 violations and extreme N=270 sample downsampling.

Methodological Limitations

- Offline Feature Extraction: CICFlowMeter calculates statistical aggregations over completed flows; real-time extraction introduces per-packet buffering overhead.
- Static Thresholding: Binary 0.5 probability thresholding fails under dynamic adversarial packet pacing.
- Synthetic Lab Traffic: CICDDoS2019 was generated in an emulated enterprise testbed, which lacks dynamic background human traffic variability.

Operational Threats to Validity

- The Base-Rate Fallacy: In Day 2, Benign traffic is only 0.28% of total traffic. An IDS with 98.7% accuracy and 1.3% False Positive Rate will trigger over 150,000 false alarms per 10 million flows!
- Shortcut Learning Risk: Classifiers remain vulnerable if attackers randomize destination ports.
- Concept Drift: Changing attack protocols requires automated retraining pipelines.

CONCLUSION & ROADMAP

Conclusions, Recommendations for Future Work & Defense Summary

1. Methodological Rigor Matters

Standard random splitting (Setup B) yields artificial 99.9% scores due to session leakage. Cross-day evaluation (Setup C) is mandatory to establish authentic real-world performance (~98%).

2. Disproved Published Literature Myths

Ahmad et al. published arithmetic F1 errors, evaluated only 270 samples, and claimed an impossible 3-7% streaming accuracy boost. We have empirically proven a 100.00% streaming match.

3. Tree Ensembles Lead in Big Data Detection

XGBoost (98.50%) and Random Forest (98.70%) achieve production-grade detection under strict temporal shifts while detecting 99.52% of unseen zero-day attacks.

4. Infrastructure Serves Scalability, Not Accuracy

Apache Kafka and Spark Streaming provide essential sub-35ms latency and horizontal scaling (14,000+ flows/s), but mathematical classification accuracy is determined by feature representations.

ROADMAP FOR FUTURE RESEARCH

- Cost-Sensitive Thresholding: Optimize ROC thresholds to suppress False Positive Rates below 0.01% for SOC usability.
- Protocol-Agnostic Payloads: Develop behavioral packet length entropy metrics that completely eliminate port shortcut reliance.
- Adaptive Online Learning: Implement Kafka streaming feedback loops that incrementally update XGBoost trees as concept drift occurs.
- Hardware-Accelerated Line-Rate: Benchmark DPDK and eBPF in-kernel flow parsing paired with Spark.

QUESTIONS & DISCUSSION

Thank you for your time.
Contact: Graduate Research Group
Code Repository: Full Colab & PySpark Pipeline